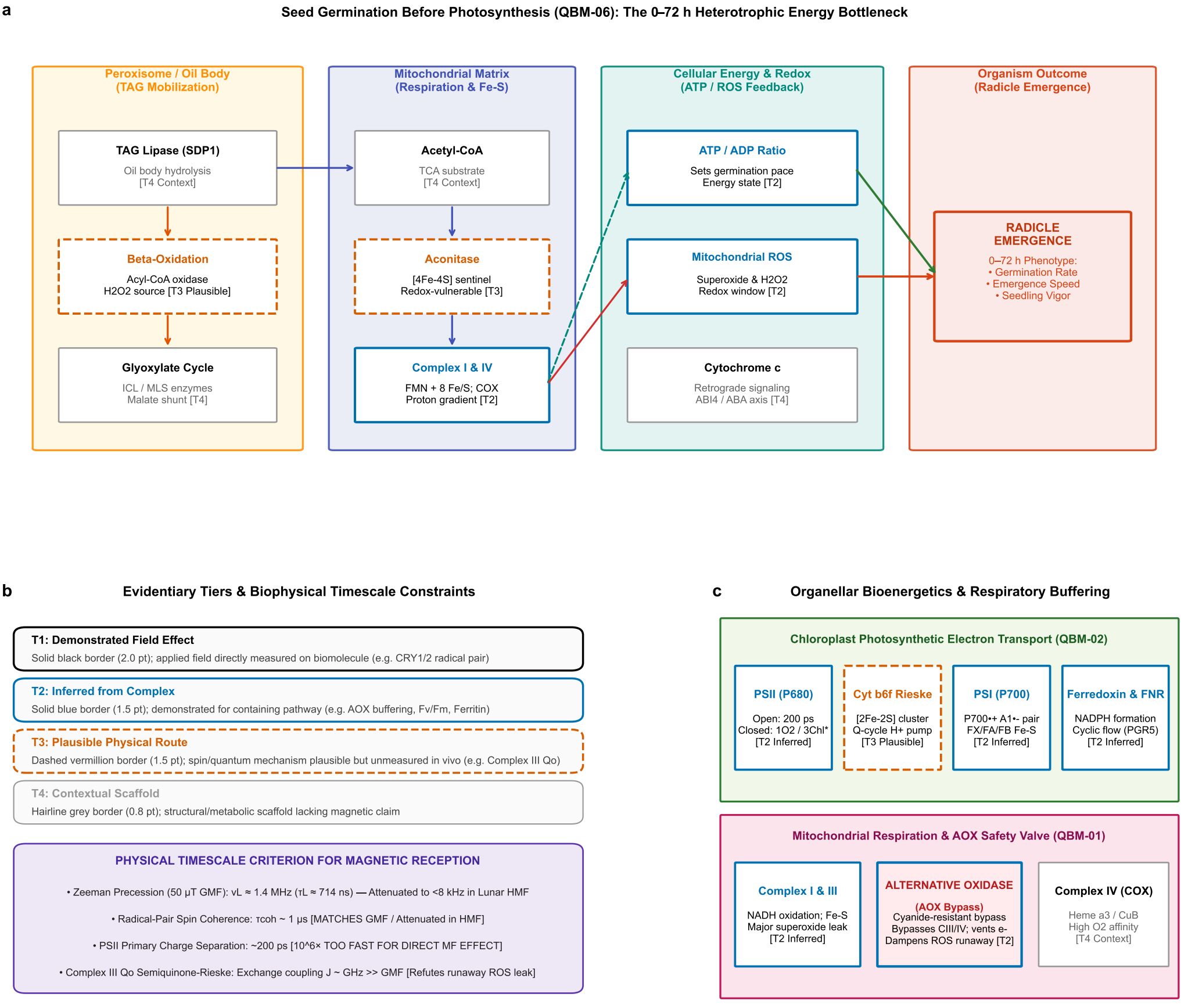
a
Seed Germination Before Photosynthesis (QBM-06): The 0–72 h Heterotrophic Energy Bottleneck
Peroxisome / Oil Body (TAG Mobilization)
TAG Lipase (SDP1)
Oil body hydrolysis [T4 Context]
Beta-Oxidation
Acyl-CoA oxidase H2O2 source [T3 Plausible]
Glyoxylate Cycle
ICL / MLS enzymes Malate shunt [T4]
Mitochondrial Matrix (Respiration & Fe-S)
Acetyl-CoA
TCA substrate [T4 Context]
Aconitase
[4Fe-4S] sentinel Redox-vulnerable [T3]
Complex I & IV
FMN + 8 Fe/S; COX Proton gradient [T2]
Cellular Energy & Redox (ATP / ROS Feedback)
ATP / ADP Ratio
Sets germination pace Energy state [T2]
Mitochondrial ROS
Superoxide & H2O2 Redox window [T2]
Cytochrome c
Retrograde signaling ABI4 / ABA axis [T4]
Organism Outcome (Radicle Emergence)
RADICLE EMERGENCE
0–72 h Phenotype:
• Germination Rate
• Emergence Speed
• Seedling Vigor
b
Evidentiary Tiers & Biophysical Timescale Constraints
T1: Demonstrated Field Effect
Solid black border (2.0 pt); applied field directly measured on biomolecule (e.g. CRY1/2 radical pair)
T2: Inferred from Complex
Solid blue border (1.5 pt); demonstrated for containing pathway (e.g. AOX buffering, Fv/Fm, Ferritin)
T3: Plausible Physical Route
Dashed vermillion border (1.5 pt); spin/quantum mechanism plausible but unmeasured in vivo (e.g. Complex III Qo)
T4: Contextual Scaffold
Hairline grey border (0.8 pt); structural/metabolic scaffold lacking magnetic claim
PHYSICAL TIMESCALE CRITERION FOR MAGNETIC RECEPTION
• Zeeman Precession (50 μT GMF): νL ≈ 1.4 MHz (τL ≈ 714 ns) — Attenuated to <8 kHz in Lunar HMF
• Radical-Pair Spin Coherence: τcoh ~ 1 μs [MATCHES GMF / Attenuated in HMF]
• PSII Primary Charge Separation: ~200 ps [10^6× TOO FAST FOR DIRECT MF EFFECT]
• Complex III Qo Semiquinone-Rieske: Exchange coupling J ~ GHz >> GMF [Refutes runaway ROS leak]
c
Organellar Bioenergetics & Respiratory Buffering
Chloroplast Photosynthetic Electron Transport (QBM-02)
PSII (P680)
Open: 200 ps Closed: 1O2 / 3Chl* [T2 Inferred]
Cyt b6f Rieske
[2Fe-2S] cluster Q-cycle H+ pump [T3 Plausible]
PSI (P700)
P700•+ A1•- pair FX/FA/FB Fe-S [T2 Inferred]
Ferredoxin & FNR
NADPH formation Cyclic flow (PGR5) [T2 Inferred]
Mitochondrial Respiration & AOX Safety Valve (QBM-01)
Complex I & III
NADH oxidation; Fe-S Major superoxide leak [T2 Inferred]
ALTERNATIVE OXIDASE
(AOX Bypass)
Cyanide-resistant bypass Bypasses CIII/IV; vents e- Dampens ROS runaway [T2]
Complex IV (COX)
Heme a3 / CuB High O2 affinity [T4 Context]